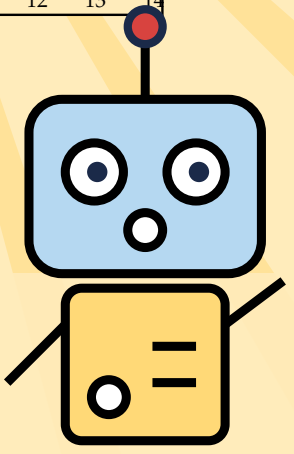
SCIENCE
Book 1

The Ruler Is Never Exact





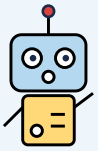
Measure it. Round it. Guess it. Check it.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

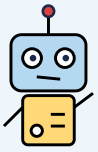


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop except, maybe, a bag of dried beans.
- **What did we keep?** After you simplify something, such as rounding a number or averaging many readings, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 44.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I measured my own foot and wrote down 25. Twenty-five what? Hold that thought. Chapter 1 will explain why I should be embarrassed.

In this book

1	Two Careful People, Two Answers	5
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⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school.

- a ruler marked in centimetres and millimetres
- a second, different ruler
- a piece of string, about 1 metre
- a pencil and a notebook
- a clock, watch or timer that shows seconds
- a small cup
- a jar or tub, bigger than the cup
- a cup of dried beans or rice
- scrap paper
- a calculator (optional)

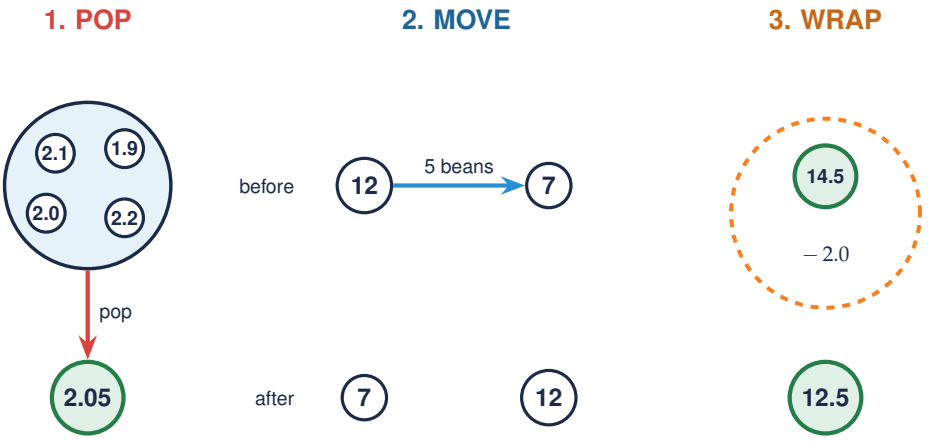
■ **Adult help:** Dried beans and rice are a choking risk for little children and pets, so keep them off the floor and put them away after each try. No one should eat the beans from the experiments. Everything else in this book is safe to do at a desk or table.

Units in this book

We use the metric system: millimetres (mm), centimetres (cm), metres (m), grams (g) and seconds (s). There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.



Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

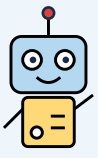
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A measurement is a circle with a note



Inside the circle there are always two things: the **number** and a **note on how it was taken**. A number with no note is a number we cannot check.



A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Two Careful People, Two Answers

Why the same desk can have two different lengths

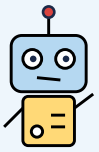
Mia and Omar share a desk at school. On Monday their teacher gives each of them the same 30 cm ruler and asks them to measure the desk, edge to edge.

Mia writes **62 cm**. Omar writes **61 cm**.

Both of them went slowly. Both of them have good eyes. Neither one is careless. So who is right?

The surprising answer is that it may be *both*, and that the desk has no single, perfect length for anyone to find. Here are four reasons the numbers can differ, and not one of them is a mistake.

1. **The ruler has to be moved.** A 30 cm ruler cannot reach across a desk in one go. Each time it is slid along, there can be a gap or an overlap about as thin as a pencil line.
2. **The edges are not sharp.** A desk corner is a little rounded. Where does the desk "start"? That is a judgement.
3. **Between two marks, you guess.** A ruler's smallest marks are 1 mm apart. If the desk ends between two marks, the reader must choose one.
4. **The ruler itself may start oddly.** Some rulers have a small gap before the zero mark, or a worn end.



My hand span is 11 cm. This desk is 6 of my hand spans. So it is 66 cm, and it is also 6 of *your* hand spans. Oh, that cannot be right. Can you see why?

A measurement is more than a number

Look at Mia's and Omar's answers as circles. Each circle holds the number and a note on how it was taken.



Two circles, two notes. With the notes, we can start to see *why* they differ.

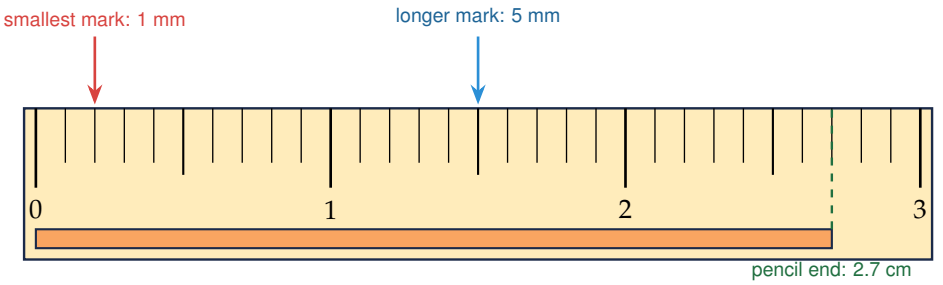
When we write a measurement without its note, we throw away the part that lets someone else check it. A scientist always keeps the note.

What did we keep?

Mia and Omar each wrote a length in centimetres. **We kept:** the number and the unit. **We set aside:** the tiny gaps, overlaps and judgements made along the way. That is why we write a *note on how it was taken*.

A closer look at a ruler

Before we go on, look hard at your ruler. Here is a piece of one, much bigger than life.



Between the 0 and the 1 there are ten small spaces, and each one is 1 millimetre. So 1 cm is 10 mm. The pencil end above lines up with the seventh small mark after the 2, so it is 2 cm and 7 mm long, which we write as **2.7 cm**.

Try it: Ruler detective

Look closely at your own ruler and fill in the lines.

(a) How many small spaces are there between the 0 and the 1? _____

(b) Is there a gap before the 0, or does the ruler start right at its edge? _____

(c) How many millimetres are there from the 2 to the 3? _____ mm

(d) Which units are written on the ruler? _____

Why we agreed on units

Long ago, people measured with their own bodies. A *hand* was a hand, a *foot* was a foot, a *cubit* was the length of a forearm. That was handy, but a big person's foot is not a small person's foot. Two shopkeepers using "one foot" could mean different amounts.

So people agreed on shared units. Today the **metre** is defined using the speed of light, which means a metre means the same thing in every country. A **centimetre** is one hundredth of a metre, and a **millimetre** is one tenth of a centimetre.

Your hand span is still useful. It tells you roughly how big something is when you have no ruler. It is just not a unit a shop could use.

Try it: Measure one desk three ways

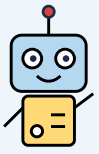
You need: a ruler, a piece of string, a pencil, and a friend. Use a table or desk about 1 metre long.

1. **Guess first.** Write your guess for the length in the table below, before you measure.
2. **Ruler.** Put a pencil dot at the end of the ruler each time you slide it along. Add up the pieces. Write down how many times you moved the ruler.
3. **Hand span.** Spread your hand wide and walk it along the desk, thumb tip to little-finger tip. Count your spans.
4. **String.** Lay the string along the desk and pinch where the desk ends. Straighten the string and measure it against the ruler.
5. **Swap.** Ask a friend to measure the same desk. Write their numbers in the second column.

	My result	My friend's	Note on how it was taken
Guess			
Ruler (cm)			
Hand spans			
String (cm)			

What to notice

Did your ruler and your string agree? Did your friend get the same number? If you were within 2 or 3 cm of each other, you did well. If you were far apart, look at the notes. A gap in the ruler path or a stretched string is usually the reason.



Hand spans are great for guessing but poor for deals. If you trade a desk to a friend and say "6 spans", ask whose spans!

Measure yourself

Your body gave people their first units. Now you can find out how big yours are.

Try it: Body measures

Measure each of these on yourself, in centimetres. Then ask a grown-up to measure theirs.

	Me (cm)	A grown-up (cm)
Hand span, thumb tip to little finger tip		
Foot, heel to longest toe		
Forearm, elbow to the tip of the middle finger (a *cubit*)		
One big step, heel to heel		

If you and the grown-up each measured the same desk in hand spans, who would get the bigger number? Explain.

_____

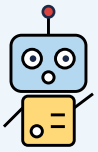
When the ruler lies a little

Mia's 62 cm and Omar's 61 cm might both be fair readings of the same desk. A 1 cm difference over about 60 cm is small. What matters is that the note is written down and that anyone who cares can try again.

Here is a simple habit that scientists use all the time:

Write the number. Write the unit. Write how you took it.

If a friend tells you a length is "21" and nothing else, you cannot tell whether that is 21 cm, 21 mm, or 21 hand spans. A number alone is a half-finished answer.



Remember my foot, from page 2? I wrote "25". I should have written "25 cm, ruler, heel to toe, read to the nearest cm." A number without its unit is like a robot without its antenna.

Puzzle 1.1: Whose report is clearest?

Three students measured the same book. Read their reports.

A "The book is 21."

B "The book is 21.2 cm long. I used a 30 cm ruler marked in mm, edge to edge, and read to the nearest mm."

C "The book is about 20 cm long."

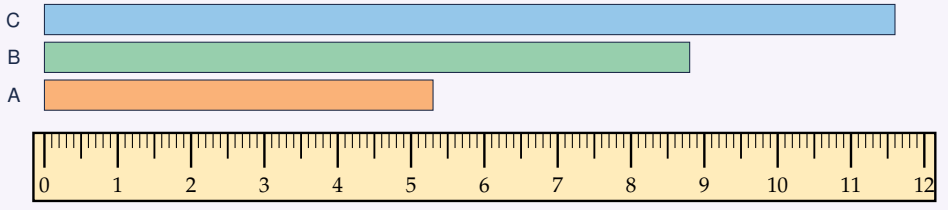
(a) Which report hides how the book was measured? _____

(b) Which report tells you the most? _____

(c) What is missing from report A besides the note? _____

Puzzle 1.2: Read the pencils

Each pencil starts at the zero mark. Read the length where each pencil ends, to the nearest millimetre.



A _____ cm B _____ cm C _____ cm

Puzzle 1.3: Odd one out

Four friends measured the same pencil. Each used a ruler marked in mm.

Ana: 17.4 cm Ben: 17.5 cm Cleo: 17.4 cm Dev: 18.4 cm

Dev's number is not like the others. Write one way Dev may have measured that gives 18.4 cm. How could Dev check?

_____

Puzzle 1.4: Match the tool

Choose the best tool for each job. Tools: **mm ruler**, **30 cm ruler for a desk**, **long tape measure**.

(a) the width of a dried bean _____

(b) the length of a classroom _____

(c) the height of a door _____

Puzzle 1.5: Treasure routes

Two routes lead from the start S to the treasure T. The lengths are in centimetres. The map is *not* drawn to scale, so trust the numbers.



(a) Length of Route 1: _____ cm (b) Length of Route 2: _____ cm
(c) Which is shorter, and by how many millimetres? _____

Puzzle 1.6: Same length, different units

Circle the longer one in each pair.

(a) 25 mm or 2.4 cm

(b) 0.8 m or 75 cm

(c) 3 cm or 28 mm

2 Round on Purpose

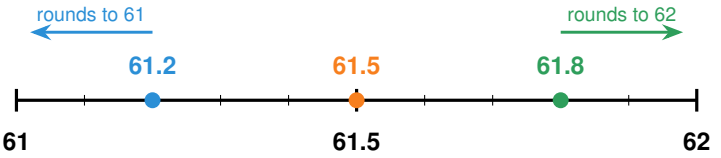
Choosing a number that says no more than you know

Mia's pencil ends exactly halfway between the 17 and 18 marks on her ruler. The ruler has no mark there. She has to choose.

Choosing the nearest mark is called **rounding**. It is not cheating, and it is not sloppy. It is a way of saying: *this is as close as my ruler lets me get.*

Which mark is nearest?

Every reading sits somewhere between two marks. Rounding picks the nearer one. Look at a number line for lengths between 61 and 62 cm.



61.2 is closer to 61, so it rounds *down* to 61. 61.8 is closer to 62, so it rounds *up*. Exactly halfway, at 61.5, we have to agree on a rule. In this book, **halves round up**, so 61.5 becomes 62.

The rounding rule

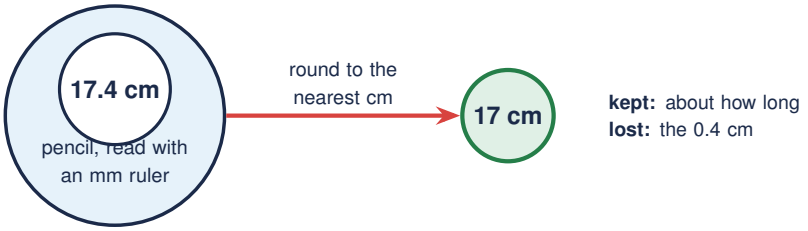
Look at the digit just after the place you want to keep.

0, 1, 2, 3, 4: the number stays. **5, 6, 7, 8, 9:** the number goes up by one.

Example: 38.4 to the nearest whole number is 38. 38.5 is 39. 38.6 is 39.

Rounding is a pop

When we round 17.4 cm to 17 cm, we squash a number into a simpler one. That is a pop, so we must say what we kept and what we lost.



What makes this correction trustworthy?

What we checked. We compared 17.4 with the rounded 17.

What this check gives us reason to trust. A number rounded to the nearest whole is never more than half a unit away from the original. The gap here is 0.4 cm, which is less than 0.5.

What it does not show. Whether the 17.4 reading itself was right, and whether adding up many rounded numbers stays close to the true total. It may not. You will see why in a moment.

Rounding to the nearest ten

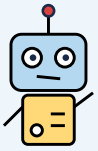
You can round to bigger steps too. To round to the nearest ten, look at the ones digit: 34 becomes 30, and 35 becomes 40. People round a distance of 487 m to "about 490 m" or "about 500 m", depending on how much detail the listener needs.

Rounding in the news

Suppose a town has exactly 12 482 people on counting day. Here is what different people might write. Each one is honest, as long as it says "about".

Rounded to the nearest	Number	Who might use it
ten	12 480	the town hall
hundred	12 500	the local paper
thousand	12 000	a national newspaper
ten thousand	10 000	a map of a whole country

Every step up throws away more detail, but each one keeps the answer to "about how big?" for a different audience.



The newspaper said 12 000 people live in the town. So the real number is exactly 12 000. I am an expert reader of newspapers. Or am I?

Try it: Round and add

You need: a ruler marked in mm, and five small things such as a pencil, an eraser, a paper clip, a key and a crayon.

1. Measure each thing to the nearest millimetre, in centimetres (for example 17.4 cm). Write each in the first column.
2. Round each length to the nearest centimetre. Write it in the second column.

3. Add the first column. Then add the second column.
4. Compare the two totals.

Thing	To the nearest mm (cm)	Rounded to the nearest cm
Total		

Practice numbers (made up for practice, not from a real measurement)

Pencil 17.4, eraser 5.2, paper clip 3.4, key 5.7, crayon 9.1.
Exact total: $17.4 + 5.2 + 3.4 + 5.7 + 9.1 = 40.8$ cm. **Rounded to the nearest cm:** 41.
Round each first: $17 + 5 + 3 + 6 + 9 = 40$. The total of the rounded numbers is *off by 1* from the rounded total of the exact ones.

Each rounding threw away a little bit, and the little bits added up. That is why careful people round *at the end*, not at every step.

What did we keep?

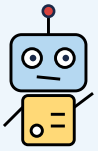
When we rounded each of the five lengths, **we kept** _____ **and we set aside** _____.

How many digits are worth writing?

Here is a puzzle that fools many grown-ups. You cut a 17 cm piece of string into 3 equal pieces. How long is each piece?

Your calculator says $17 \div 3 = 5.6666667$.

But your ruler only has marks every millimetre. The very last digit of 5.6666667 is about a millionth of a millimetre! No ruler can show that, so writing it pretends you know something you cannot know. The honest answer is **5.7 cm**.



The calculator gave me 5.6666667, so my string piece is 5.6666667 cm. More digits means more exact. Right? I think that is wrong. Can you say why?

Do not write more digits than your tool can honestly give.

Puzzle 2.1: Round the number

Round each number. Remember: halves round up.

To the nearest whole number		To the nearest ten	
(a) 23.4	_____	(f) 45.2	_____
(b) 23.5	_____	(g) 81	_____
(c) 7.49	_____	(h) 135	_____
(d) 99.6	_____		
(e) 12.51	_____		

Puzzle 2.2: Sensible digits

Your calculator shows a long answer. Your ruler is marked in millimetres. Write each answer in centimetres, to the nearest millimetre.

(a) $17 \div 3 = 5.6666667$ write: _____ cm

(b) $33 \div 7 = 4.7142857$ write: _____ cm

(c) $50 \div 8 = 6.25$ write: _____ cm

Puzzle 2.3: Which rounding is wrong?

Four people rounded these lengths to the nearest centimetre. Only one rounding is wrong.

$17.4 \to 17$ $8.5 \to 8$ $26.7 \to 27$ $3.2 \to 3$

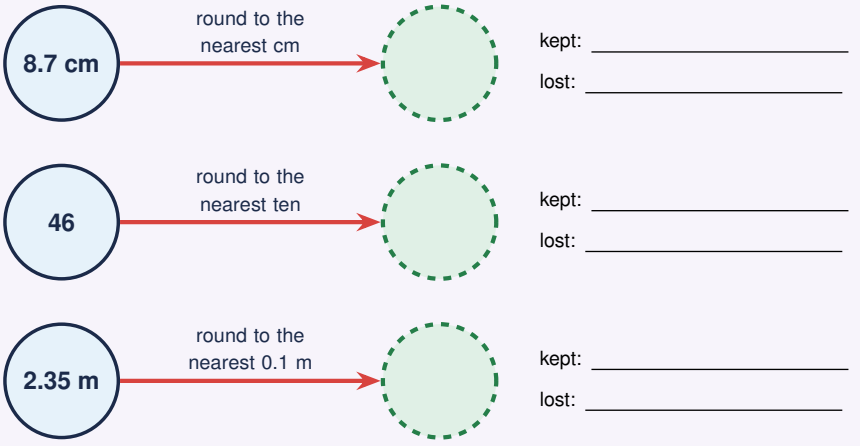
Which one is wrong? _____ What should it be? _____

Puzzle 2.4: Ribbons

Three pieces of ribbon are each 1.4 cm long.
(a) What is their total length, exactly? _____ cm
(b) Round each piece to the nearest centimetre first, then add. What do you get? _____ cm
(c) The two answers are different. Which total is closer to the truth, and why?
_____

Puzzle 2.5: Draw the pop

Fill in the empty circle for each pop. Then write what was kept and what was lost.



Puzzle 2.6: Rounding maze

Start at the top left. Move only **right** or **down**, one square at a time. You may step only on a number that rounds to **50** when rounded to the nearest ten. Reach the bottom right.

START

47	61	38
52	54	29
33	49	45

END

3 Guess First

Estimating before you measure, then correcting

Before you measure anything, you can guess. A guess is not a wild shot. It is a quick, rough answer that tells you what size of answer to expect.

Why bother? Suppose you measure a pencil and your ruler says 1.7 m. If you had guessed first, you would know at once that something is off, because a pencil is not as long as a person's arm. A guess is a smoke alarm for silly slips.

Try it: Guess, then measure

Pick three things in the room. Write a guess for each length in centimetres. Then measure and see how far off you were.

Thing	My guess (cm)	Measured (cm)	Off by (guess − measured)

A negative "off by" means you guessed too small. A positive one means you guessed too big. Over time, most people's guesses get better, because they learn what 10 cm, 30 cm or 1 m looks like.

Benchmarks: rulers you carry in your head

A **benchmark** is a thing whose size you know well. When you have no ruler, you can compare an unknown length to a benchmark.

Benchmark	About how big
A bank card or bus pass card	8.6 cm by 5.4 cm
A sheet of letter paper	21.6 cm by 27.9 cm (A4 paper: 21.0 cm by 29.7 cm)
A new, unsharpened pencil	about 19 cm
A room door	about 2 m high

Real things vary a little, so check yours with a ruler once and remember the number.

Try it: Estimate with a sheet of paper

Lay sheets of paper end to end along a desk, long side along the desk. Count how many sheets and part-sheets it takes. Multiply by 27.9 cm if you use letter paper, or by 29.7 cm if you use A4. Write the estimate, then check with your ruler.

Sheets: _____ Estimate: _____ cm Ruler: _____ cm

Beans in a jar

Now for a bigger guess. How many beans are in a jar? You could count them all, but if the jar is big that takes ages. Instead we can **estimate**: count a small scoop, and use the scoop to work out the whole jar.

Try it: Estimate the beans

You need: a cup of dried beans, a small cup, a jar or tub, a spoon or ruler to level the cup.

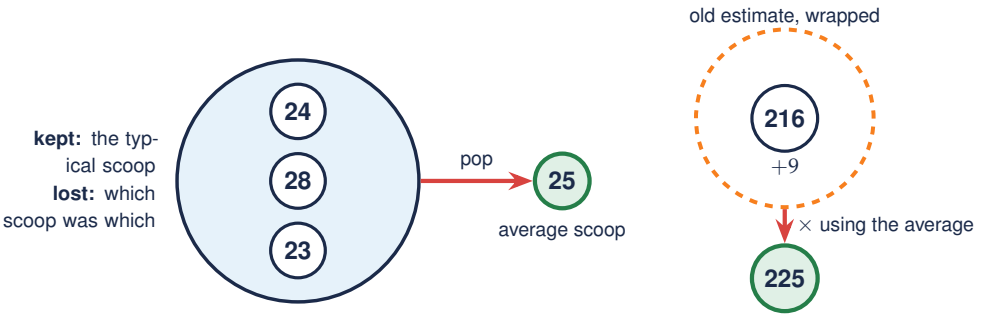
1. Fill the small cup with beans and level the top. Tip them out and **count** them. That is one scoop.
2. Pour beans from the small cup into the jar, one level cup at a time, until the jar is full. Count the cups.
3. Multiply: beans in one scoop $\times$ number of cups $=$ **estimate**.
4. Now take two more scoops and count each. Average the three scoops (add them, divide by 3).
5. Multiply the **average** by the number of cups. This is your corrected estimate.
6. Only now, count every bean in the jar and compare.

■ **Adult help:** Beans can be a choking hazard for little children and pets. Count them at a table and put them away afterwards. Do not eat them.

Practice numbers

Made-up numbers for practice, not from a real jar

One scoop had 24 beans. The jar took **9** level cups.
First estimate: $24 \times 9 = 216$.
Two more scoops: 28 and 23. Three scoops: 24, 28, 23. Total $24 + 28 + 23 = 75$. **Average:** $75 \div 3 = 25$.
Corrected estimate: $25 \times 9 = 225$. The correction is $+9$ beans.
Real count: 231. The corrected estimate was off by $231 - 225 = 6$.



What makes this correction trustworthy?

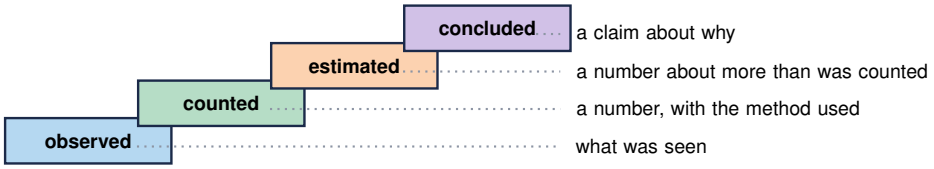
What we checked. We took three scoops instead of one, and we compared the corrected estimate, 225, with a full count, 231.

What this check gives us reason to trust. The average of three scoops is steadier than one scoop, and the final answer was only 6 away from the real count, which is under 3 beans in every hundred.

What it does not show. That a different jar, a different kind of bean or a heaped cup would give the same result. One careful jar shows that the method *can* work, not that it always does.

The ladder from seen to concluded

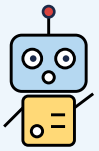
Scientists keep four kinds of statement apart. We will use this ladder in every science book.



In the bean jar, "there were 24 beans in my first scoop" is a **count**. "There are about 225 beans in the jar" is an **estimate**, because we counted a few and reasoned about all. Notice the word "about". We do not say there are exactly 225.

What did we keep?

We used three scoops to estimate the whole jar. **We kept** _____, **and we set aside** _____.



Three scoops gave 24, 28 and 23, and the jar holds 9 cups. So $24 + 28 + 23 = 75$ beans in the jar! That was easy. Why do you think the real count was so much bigger?

No peeking

Do not count the whole jar first and then "estimate". An estimate made after peeking at the answer proves nothing. Write the estimate down *before* the full count, so the correction is honest.

Pacing it out

You do not need a tape measure to estimate a long distance. You can walk it. First find out how long *your* walking step is, then count steps.

Try it: Pace a hallway

You need: a long tape measure or a long piece of string and a ruler, plus a hallway or a path.

1. Walk 10 normal steps in a straight line. Measure the distance. Divide by 10 to get the length of one pace.
2. Do it three times and average the three distances. Your pace length is that average divided by 10.
3. Walk the hallway and count your paces. Multiply the paces by your pace length.
4. Measure the hallway with the tape and compare.

Practice numbers, not from a real hallway

First try. 10 paces measured 7.2 m, so one pace is 0.72 m. The hallway took 31 paces. $31 \times 0.72 = 22.32$, so **about 22 m**.
Corrected. Two more tries: 7.7 m and 7.4 m. Total $7.2 + 7.7 + 7.4 = 22.3$, and $22.3 \div 3 \approx 7.43$ m for 10 paces. One pace is 0.743 m. $31 \times 0.743 \approx 23.0$, so **about 23 m**.
Tape measure. 22.9 m. The first estimate was off by about 0.6 m. The corrected one was off by about 0.1 m.

What did we keep?

When we walked a hallway to measure it, **we kept** _____ **and we set aside** _____.

Puzzle 3.1: Which guess was closer?

A jar really had 231 beans. Three people guessed.

Priya: 190 Sam: 260 Tomas: 225

For each, work out how far off the guess was.
Priya _____ Sam _____ Tomas _____
Whose guess was closest? _____

Puzzle 3.2: Fill the jar

Four scoops of lentils had 18, 20, 19 and 19 lentils. The jar holds 12 scoops.
(a) Total of the four scoops: _____
(b) Average scoop: _____
(c) Estimate for the jar: _____

Puzzle 3.3: Scoop detective

Four scoops of beans were counted: **24, 28, 23 and 52**.
One count looks strange. Which one? _____
Write one way it might have happened, and what you would do about it.
_____

Puzzle 3.4: Smoke alarm

Each measurement has a slip. Use a quick guess to spot it.

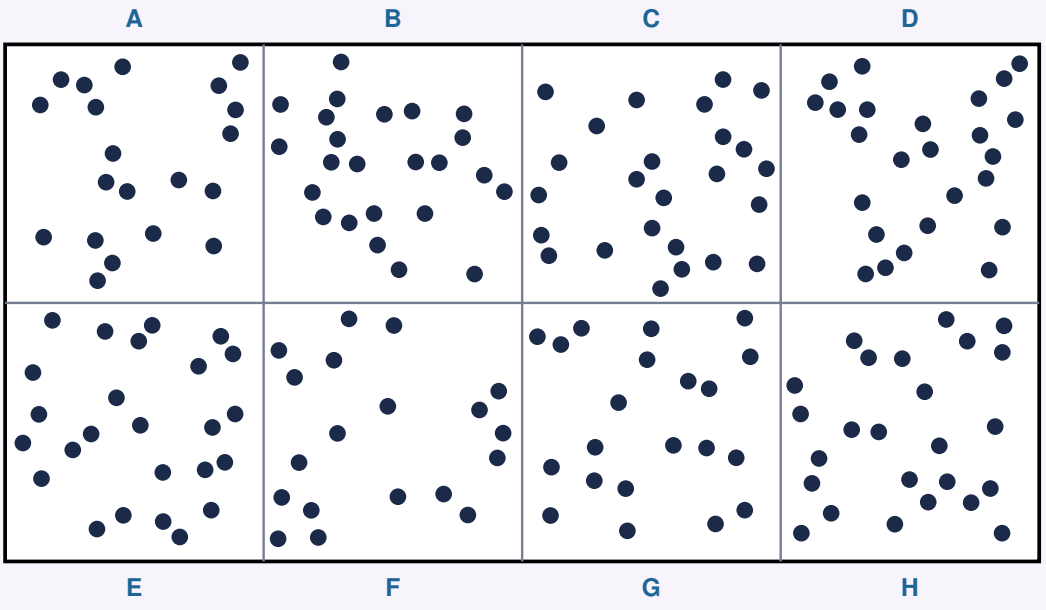
(a) "My pencil is 175 cm long." Slip? _____

(b) "This classroom door is 2 mm tall." Slip? _____

(c) "My shoe is 25 m long." Slip? _____

Puzzle 3.5: Dots in a field

There are far too many dots to count one by one. The field is split into 8 equal squares, A to H. A to D are along the top and E to H along the bottom.



(a) Count the dots in squares **B**, **C** and **F**. B: _____ C: _____ F: _____
(b) Average per square: _____ (c) Estimate for the whole field (average × 8): _____
(d) Now count every dot if you dare. Was your estimate close? _____

4 Ten Readings, One Answer

Why careful scientists measure again and again

Try this: hold a sheet of paper flat, let it go, and time how long it takes to reach the floor. Do it again. Did you get the same time?

Almost nobody does. Your thumb is sometimes quick on the stopwatch and sometimes slow. The paper wobbles a different way each time. Every reading is a little different, even though you did everything carefully.

Scientists do not get upset about this. They plan for it. If one reading wanders a little, many readings tend to wander in different directions and cancel out. So they measure again and again, and they report the **average** (also called the **mean**) together with the **spread**.

Try it: Time ten falls

You need: a clock or timer that shows seconds, a sheet of scrap paper, and a pencil. A helper makes it easier.

1. Stand on the floor and hold the sheet flat, as high as you can reach safely. Do not stand on a chair.
2. Let go and time the fall, to the nearest tenth of a second.
3. Write each time in the table. Do this ten times, keeping the sheet and your starting height the same.
4. Find the **total** of the ten times. Divide by 10. That is the **average**.
5. Find the **spread**: the biggest time minus the smallest time.

1	2	3	4	5	6	7	8	9	10

Average: _____ s Spread: _____ s

■ **Adult help:** Stand on the floor, not on furniture. Do this where no one can trip over a falling sheet.

Practice numbers

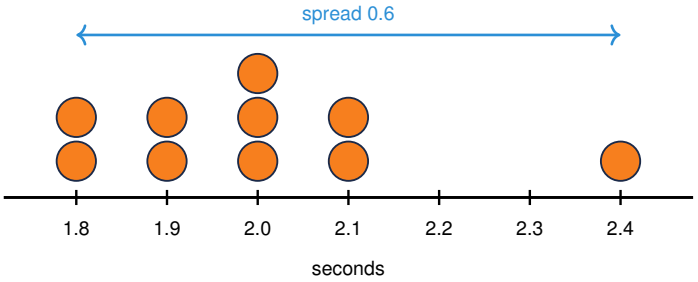
Here are ten fall times from a made-up practice run. Each is in seconds.

Practice numbers, not from a real experiment

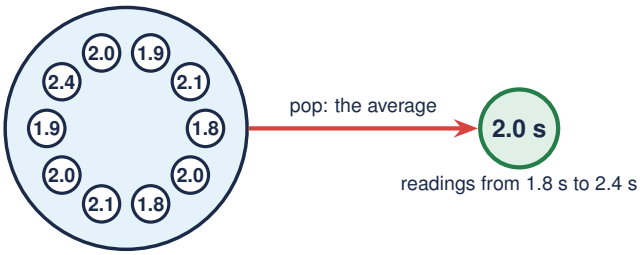
1.8 2.1 1.9 2.0 2.4 1.9 2.0 2.1 1.8 2.0

Total: $1.8 + 2.1 + 1.9 + 2.0 + 2.4 + 1.9 + 2.0 + 2.1 + 1.8 + 2.0 = 20.0$.
Average: $20.0 \div 10 = 2.0$ seconds.
Spread: biggest minus smallest $= 2.4 - 1.8 = 0.6$ seconds.

A **dot plot** shows the readings at a glance. Each dot is one reading. Tall stacks are common times.



Ten circles become one



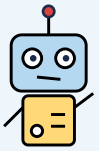
What did we keep?

The average and the spread tell us a lot. **We kept** _____ **and we set aside** _____.

The odd one out

In the practice run, one reading, 2.4, stands apart from the others. Do not delete it just because it is awkward! Ask why it happened. Maybe you started the clock late. If there is a good reason, write the reason down.

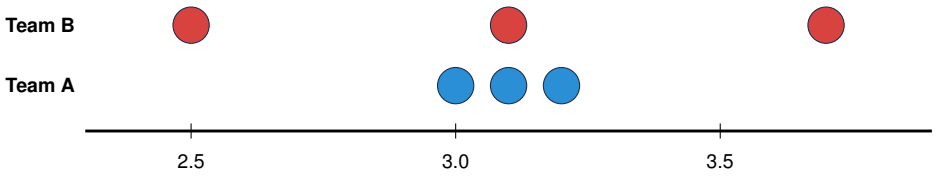
Here is what happens if you leave 2.4 out. The other nine readings total $20.0 - 2.4 = 17.6$, and $17.6 \div 9 = 1.96$, which rounds to 2.0. Leaving it out hardly changed the answer. Whenever you set a reading aside, say so in your notes.



I have a plan. I will delete every reading that I do not like, and then my results will be very tidy. A scientist would not do that, would they?

Same average, different spread

Two teams each timed a rolling ball three times. Both teams found the same average. Do their results look equally good?



Team A: 3.1, 3.0, 3.2 (average 3.1, spread 0.2). Team B: 2.5, 3.1, 3.7 (average 3.1, spread 1.2). Same average, but Team B's readings are all over the place. The spread tells you how *steady* the readings are. A small spread means the readings agree, and a big spread means they do not.

What makes this correction trustworthy?

What we checked. We measured ten times and took the average. The result was 2.0 seconds with readings from 1.8 to 2.4.

What this check gives us reason to trust. Random slips, such as pressing the button a little early or a little late, tend to cancel when you take many readings, so the average is steadier than any single one. If someone else repeats the experiment and gets a similar average, the result can be repeated.

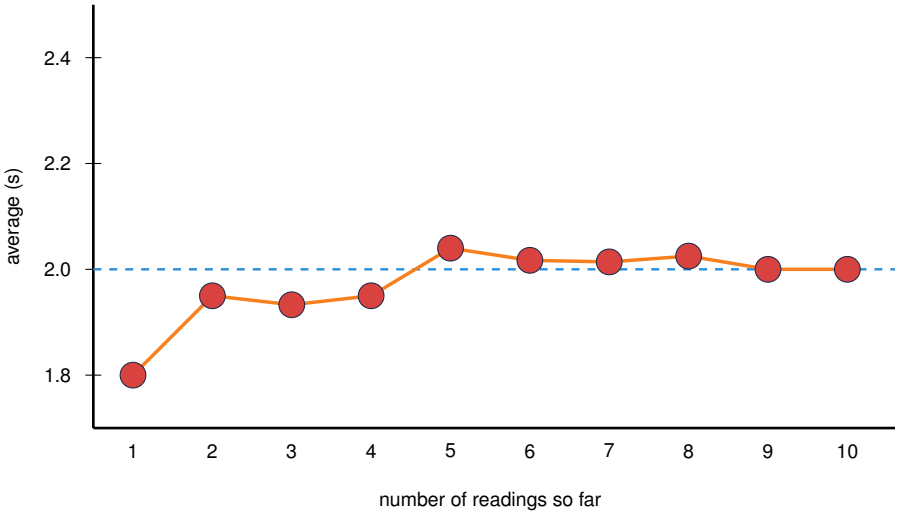
What it does not show. Whether the timer itself is right, or whether something pushes every reading the same way. A clock that runs fast makes every time too long, and averaging cannot fix that. Agreement between repeats shows that a result is *repeatable*, not that it is *true*.

Another way to take the middle

You can also find the **middle value**. Put the readings in order and pick the one in the middle. For our ten readings sorted, 1.8, 1.8, 1.9, 1.9, 2.0, 2.0, 2.0, 2.1, 2.1, 2.4, the middle two are both 2.0, so the middle value is 2.0. Here the average and the middle agree. They do not always.

Does the average settle down?

Watch the average as the readings come in. After each new reading, work out the average of all the readings so far.



The first average is just the first reading, 1.8. The wiggles get smaller as more readings are added. After about six readings the average stays within 0.03 s of 2.0. One reading can be far from the truth. An average of many is steadier.

Try it: Catch the ruler

How fast do you react? One person holds a 30 cm ruler at the top, with zero at the bottom. Another holds a thumb and finger open at the zero mark, ready to pinch. The holder lets go without warning. The catcher pinches as fast as they can. Read the mark where the ruler was caught, to the nearest centimetre. That is how far it fell.

1. Do ten catches. Write each distance in centimetres.
2. Find the total, the average and the spread.

1	2	3	4	5	6	7	8	9	10

Total: _____ cm Average: _____ cm Spread: _____ cm

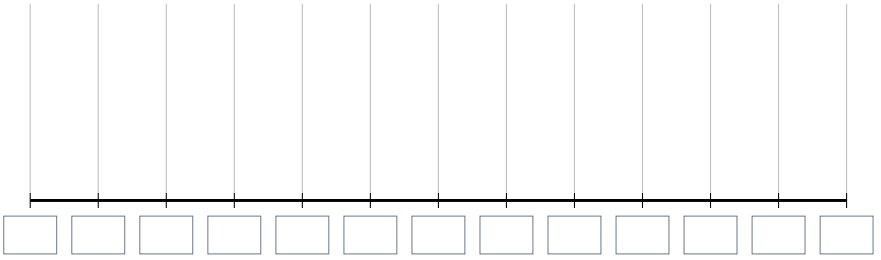
Practice numbers, not from real catches

12, 15, 11, 14, 13, 16, 12, 13, 14, 10 (cm).
Total = 130. Average = 130 ÷ 10 = 13.0 cm. Spread = 16 − 10 = 6 cm.
A ruler that falls 13 cm takes about one sixth of a second. So this catcher reacts in about

one sixth of a second. Notice that single catches range from 10 to 16 cm, so one catch alone would be a poor guide.

Make your own dot plot

Put your ten catches on this axis. Write the numbers under the marks that suit your readings, then draw one dot for each reading. Stack dots that land on the same mark.



write the numbers in the boxes

Puzzle 4.1: Find the average

Five students timed a toy car. Their times in seconds were 12, 14, 13, 15 and 11.
Total: _____ Average: _____ Spread: _____

Puzzle 4.2: Slip or real?

Five readings of a length in centimetres: 4.1, 4.0, 4.2, 4.1, 41.
(a) Which reading looks like a slip? _____
(b) What do you think went wrong?
_____
(c) If the slip is corrected, what is the average of the five readings? _____

Puzzle 4.3: Who is steadier?

Two classes each timed a pendulum swing three times. Class P: 2.9, 3.0, 3.1. Class Q: 2.2, 3.0, 3.8.
(a) Average of P: _____ Average of Q: _____
(b) Spread of P: _____ Spread of Q: _____
(c) Which class's readings agree better? _____

Puzzle 4.4: The runaway clock

Mina's kitchen clock gains 3 seconds every 100 seconds. She times a 100-second event ten times with it and gets a very steady average.
Does her steady average prove her clock is right? Explain in one sentence.
_____

Puzzle 4.5: Rounded average

A student read a length eight times, in centimetres: 3.1, 3.2, 3.1, 3.3, 3.1, 3.0, 3.2, 3.1.
(a) Total: _____ (b) Average: _____ (c) Average rounded to the nearest tenth: _____
(d) Spread: _____

5

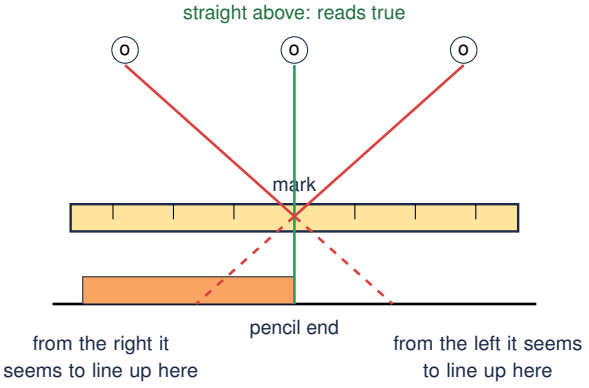
Reading the Instrument

Rulers, scales and the places where mistakes like to hide

A ruler, a scale and a stopwatch are called **instruments**. Instruments help your eyes and hands, but they bring their own ways of going wrong. This chapter is about three common ones.

1. Look straight on

Hold a ruler a little above a pencil and look at the end of the pencil from different places. The mark that seems to line up with the end changes, because the ruler is a little way above the pencil. This is called **parallax**.



The fix is easy: put your eye straight above the mark you are reading, and keep the ruler as close to the object as you can.

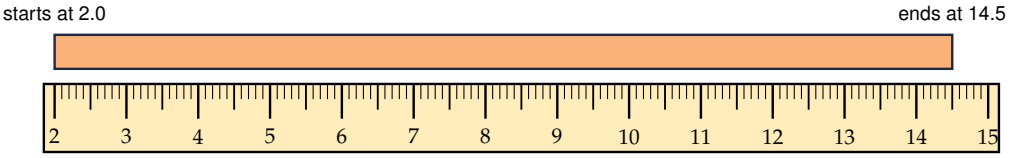
Try it: Try three viewpoints

Lay a pencil flat. Hold a ruler a little above it so the zero mark lines up with the pencil's left end. Read the right end three times: from straight above, from a little to the left, and from a little to the right. Write down the three readings. Do they match?

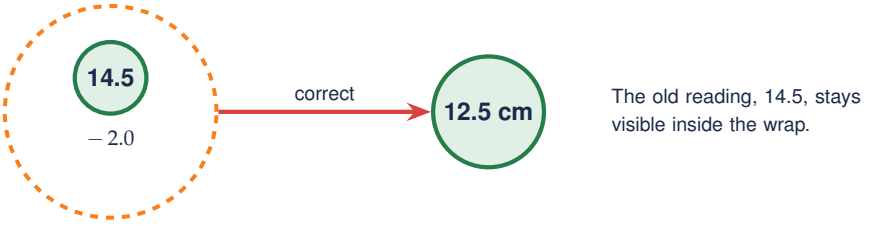
Straight above _____ cm Left _____ cm Right _____ cm

2. Where does zero start?

Many rulers start at zero right at the edge. But some have a gap before the zero, and old rulers get worn at the end. If the zero is missing, you can start from a different mark, say the 2 cm mark, and **subtract** what you skipped.



The pencil starts at 2.0 and ends at 14.5. Its length is $14.5 - 2.0 = 12.5$ cm. The reading 14.5 on its own would be wrong, and we wrap it in a correction.



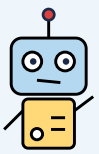
Try it: Find your ruler's zero

Look closely at the left end of your ruler. Is there a gap before the zero mark? How wide is it? Now measure a pencil twice: once with its end at zero, and once with its end at the 2 cm mark. Subtract 2. Do you get the same length?

Gap before zero: _____ mm From zero: _____ cm From 2 cm (after subtracting): _____ cm

3. A scale that is not at zero

Some instruments are off from the start. A kitchen scale might show 3 g when nothing is on it. That is called a **zero error**. If an apple makes it read 152 g, the apple is not 152 g. It is $152 - 3 = 149$ g.

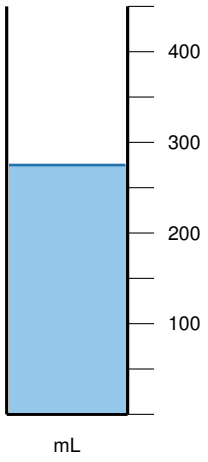


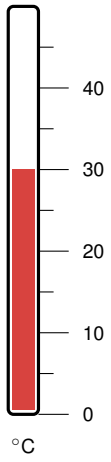
My scale showed 3 g with nothing on it. That is fine. Zero is just a feeling. I will ignore it and write 152 g for the apple. Is the apple 152 g?

■ **Adult help:** A kitchen scale is a good place to try this. An adult should help with the scale and with any food items.

4. Reading a jug and a thermometer

Not every instrument is a ruler. A measuring jug and a thermometer have marks too, and the same rule applies: read the nearest mark, and say which unit you are using. For a jug, put it on a flat table and read the level at eye height. The liquid curves up at the edges, so read the flat middle.





What do you read?

The jug has a mark every 50 mL, with the numbers on the 100s. The water sits halfway between the 250 and 300 marks, so it is 275 mL. When a level is between marks, say "about halfway between 250 and 300" or pick the nearer mark. On the thermometer, each small step is 5 degrees, and the red line is at 30. The thermometer says 30 °C.

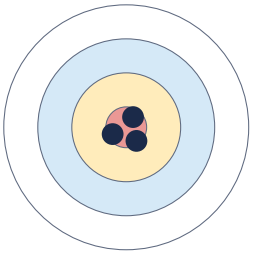
Try it: Read the jug

Pour water into a measuring jug on a flat table. Put your eyes level with the water. Read the number. Pour a little more and read again. Write each reading with its unit.

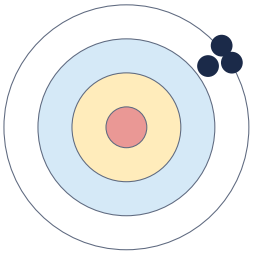
First reading _____ mL Second reading _____ mL Added _____ mL

Steady is not the same as right

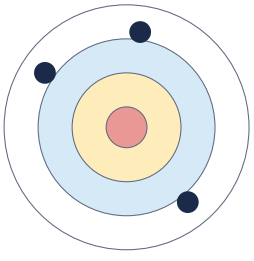
Three archers each shot three arrows at a target. The middle is the true value.



steady and right



steady but wrong



scattered but near

Steady readings that are near each other do not have to be near the truth. A scale with a zero error gives steady but wrong answers, and the readings can look perfect while all being too big. Repeating tells you about steadiness. Only checking against a better tool, or correcting a known fault, tells you about being right.

5. Two rulers are better than one

If you have two different rulers, measure the same pencil with each. If both rulers agree, that is good news. If they disagree, one of them is off, or you misread one.

Practice numbers, not from a real pencil

Ruler A: 12.5 cm. Ruler B: 12.4 cm. The difference is 0.1 cm, which is 1 mm. That is about the size of the smallest mark, so the rulers agree well.

What makes this correction trustworthy?

What we checked. We measured the same pencil with two different rulers and compared. They differed by 1 mm.

What this check gives us reason to trust. Two different rulers, read separately, agreeing to within a millimetre makes it unlikely that either one has a big fault or that we misread both in the same way.

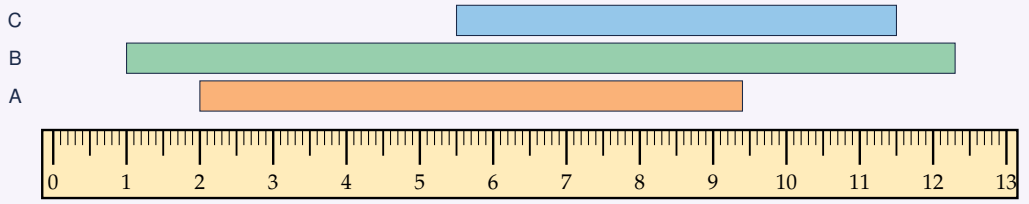
What it does not show. Whether both rulers share the same fault, for example if they came from the same maker, or whether the pencil is the length we think it is. Agreement between two tools is a good sign, not a guarantee.

What did we keep?

When we corrected a reading by subtracting a zero error, **we kept** _____ **and we set aside** _____.

Puzzle 5.1: Read the shifted pencils

Each pencil starts at a mark that is *not* zero. Read the start and the end, then subtract to find the length.



A: start _____ end _____ length _____ B: start _____ end _____ length _____

C: start _____ end _____ length _____

Puzzle 5.2: Zero error hunt

(a) A scale shows 5 g when it is empty. A potato makes it show 205 g. How heavy is the potato? _____

(b) A ruler's end is snapped off, so it starts at the 1.5 cm mark. One edge of a book lines up with 1.5 and the other edge with 22.0. How long is the book? _____ cm

Puzzle 5.3: Spot the bad habit

Four students gave advice about reading a ruler. Only one tip is bad. Which one?

A Put your eye straight above the mark.

B Line up the object with a clear mark, not with the ruler's edge.

C Read the mark from the side, at an angle, so you can see it better.

D Write down the unit.

The bad tip is _____ because _____.

Puzzle 5.4: Jugs and thermometers

(a) Three jug levels are marked at 150, 275 and 400 mL. How much water must you add to go from the first to the last? _____ mL

(b) Three thermometers read 18, 27 and 35 °C. How many degrees warmer is the warmest than the coolest? _____ °C

(c) A jug reads 275 mL, but a better jug says 250 mL. Is the first jug steady, right, or neither? Say what you know and do not know.

_____

6 Reporting What You Found

Saying what you know, and how well you know it

A measurement that stays in your head helps no one. A good report lets a stranger see what you did, so they can check your work or repeat it.

The five-line report

Every report in this series can use the same five lines.

1. **What I measured.** The thing, and the quantity (length, time, mass).
2. **How I measured it.** The instrument, the unit, how it was read, how many times.
3. **My readings.** All of them, not just the one I liked best.
4. **My answer.** The average or best value, rounded sensibly, with the spread.
5. **What I kept and what it hides.** What the answer keeps, and what it leaves out.

Here is a report built from the practice numbers in Chapter 4.

A sample report (practice numbers)

1. What I measured. The time a sheet of scrap paper took to fall to the floor, released flat from the height of my raised hand.
2. How I measured it. A stopwatch read to the nearest 0.1 s. One person timed ten falls with the same sheet.
3. My readings (s). 1.8, 2.1, 1.9, 2.0, 2.4, 1.9, 2.0, 2.1, 1.8, 2.0.
4. My answer. About 2.0 s. The readings ranged from 1.8 s to 2.4 s, a spread of 0.6 s.
5. What I kept and what it hides. The average and the spread are kept. The order of the readings, and which drop was which, are set aside. A different person, sheet or height may give a different answer.

Which rung of the ladder?

Remember the ladder from Chapter 3: observed, counted, estimated, concluded. A good report keeps them apart. Look at these four sentences from the sample report. Each one sits on a different rung.

Puzzle 6.1: Climb the ladder

Write O (observed), C (counted), E (estimated) or K (concluded) next to each sentence.

(a) "The sheet wobbled as it fell." _____

(b) "I timed ten drops." _____

(c) "A sheet like this takes about 2.0 s to fall from my reach." _____

(d) "Paper falls slowly because air pushes up on it." _____

Which of these sentences did the ten readings *not* test? _____

Five ways to check, five things they show

The trust box in this book used *repeat and compare*. The other series use other kinds of check. Each one can show something, and none of them shows everything.

Kind of check	What it can show	What it cannot show
Prove (math)	A result must hold for every case the argument covers.	That the rule fits the real world.
Compare with measurement (science)	A prediction matched what was measured this time.	That it will match every time and everywhere.
Repeat and compare (science, ecology)	A result can be repeated.	That it is true, or that the test was fair.
Check against how words are used (spelling and grammar)	A choice fits dictionaries and good usage.	That it is the only choice, or that a word's whole history is known.
Check against sources (Problem Solving)	What surviving evidence says.	That the evidence is complete.

Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is not who made it but whether it shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty science poster. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so that the bars cannot be read.

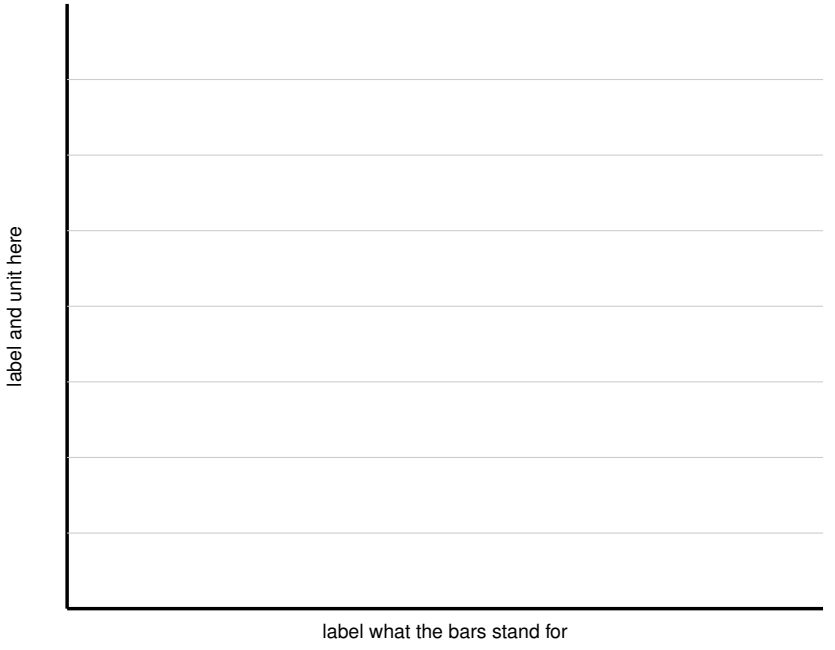
_____

(d) Which part is not relevant to measuring at all?

_____

Now you try

On a separate sheet, redraw the poster with the same eight bars. Give it a title that says what was measured, label the axes with units, pick one or two calm colours, and delete everything that does not help. Check your poster against the three words: salient, simple, relevant.



Title:

Mixed review

Each puzzle uses something from an earlier chapter.

Puzzle 6.2: Too many digits

A student writes: “My pencil is 17.43928 cm long.” The ruler is marked in millimetres. What is wrong, and what should the student write? _____

_____

Puzzle 6.3: Round, average, report

A string was measured five times, in centimetres: 24.6, 24.8, 24.7, 24.6, 24.9.

(a) Total: _____ (b) Average: _____ (c) Average to the nearest tenth: _____

(d) Spread: _____

(e) Write the answer as one honest sentence.

_____

Puzzle 6.4: Match the problem to the fix

Draw a line from each problem to the best fix.

(a) Each reading is a little different	look straight on
(b) A scale shows 4 g with nothing on it	measure many times, average
(c) The mark seems to move as you lean	round to what the tool can show
(d) The calculator shows nine digits	subtract the zero error

My measurement log

Write down things you measure. Always include the unit and the tool.

Day	What I measured	Tool	Reading and unit	Anything odd?

Your turn: write a five-line report

Choose any measurement from this book: the desk, the beans, the falling paper, the catch, the pencil. Fill in the five lines so a stranger could repeat what you did.

1. What I measured.

2. How I measured it.

3. My readings (with units).

4. My answer, rounded sensibly, with the spread.

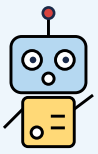
5. What I kept and what it hides.

Check your report against the trust box questions. What did you check? What does that give you reason to trust? What does it not show?

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Two answers	a desk length written as one number	the number, unit and how it was taken	small gaps, overlaps, judgements
2 Round on purpose			
3 Guess first			
4 Ten readings			
5 The instrument			
6 Reporting			



Everything you have measured in this book was a simplification of something. Now you know how to say what was kept. Next time somebody gives you a number, you will want to ask: how was it taken?

Questions I still have

Write a question about measuring that this book made you wonder about. Then write how you might find out.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on hand spans. Spans are different sizes for different people. Six of *Multix's* spans is not six of *yours*, so the desk is not "66 cm" unless the span is agreed. **Keep box.** Kept: the number and unit. Set aside: small gaps, overlaps and judgements.

1.1 (a) A hides how the book was measured. (b) B tells the most: unit, instrument, edge to edge, nearest mm. (c) A has no unit. 21 what?
1.2 A 5.3 cm, B 8.8 cm, C 11.6 cm.
1.3 Dev probably started the pencil at the 1 cm mark and read the other end at 18.4. Subtracting the 1 cm gives 17.4 cm, which fits the others. To check, measure again starting at zero, or use a second ruler.
1.4 (a) mm ruler. (b) long tape measure. (c) long tape measure.
1.5 Route 1: $3.5 + 2.0 + 4.5 + 1.5 = 11.5$ cm. Route 2: $2.5 + 3.5 + 2.0 + 3.0 = 11.0$ cm. Route 2 is shorter, by 0.5 cm, which is 5 mm.
1.6 (a) 25 mm, because 25 mm is 2.5 cm. (b) 0.8 m, because it is 80 cm. (c) 3 cm, because it is 30 mm.

Chapter 2

Multix on newspapers. No. "12 000" is a rounded figure, so the real number could be anywhere from about 11 500 to 12 499. **Multix on digits.** More digits is not more exact. The ruler can only show millimetres, so any digit beyond that is made up. **Keep box.** Kept: about how long each thing is, and the total to the nearest whole. Set aside: the tenths of a centimetre, which add up.

2.1 (a) 23 (b) 24 (c) 7 (d) 100 (e) 13 (f) 50 (g) 80 (h) 140.
2.2 (a) 5.7 cm (b) 4.7 cm (c) 6.3 cm (halves round up).
2.3 $8.5 \to 8$ is wrong. It should be 9 (halves round up).
2.4 (a) $1.4 \times 3 = 4.2$ cm. (b) $1 + 1 + 1 = 3$ cm. (c) 4.2 is closer. Each rounding loses 0.4 cm, three times over, so the pieces lose 1.2 cm together. Round at the end.
2.5 $8.7 \to 9$ (kept: about how long; lost: the 0.3). $46 \to 50$ (kept: about how big; lost: the 4). $2.35 \to 2.4$ (kept: about how long; lost: the 0.05). Other wordings are fine if they name what was kept and what was lost.
2.6 The path is 47, 52, 54, 49, 45, because each of these rounds to 50.

Chapter 3

Multix on the jar. 75 is the total of the three scoops, not the jar. First find the average scoop, $75 \div 3 = 25$, then multiply by 9 cups: $25 \times 9 = 225$. **Keep boxes.** Beans: kept the typical scoop (the average), set aside which scoop was which and how much they differed. Pacing: kept the typical pace length, set aside the small differences between paces.

3.1 Priya 41 off, Sam 29 off, Tomas 6 off. Tomas was closest.
3.2 (a) $18 + 20 + 19 + 19 = 76$. (b) $76 \div 4 = 19$. (c) $19 \times 12 = 228$.

3.3 52. It is about double the others. Perhaps two scoops were counted together, or the cup was heaped. Recount that scoop, and write down what you find.
3.4 (a) A pencil is about 17.5 cm, so 175 has a decimal slip. (b) A door is about 2 m, not 2 mm: a unit slip. (c) A shoe is about 25 cm, not 25 m: a unit slip.
3.5 The practice field has B = 24, C = 25, F = 19. Total 68, average $68 \div 3 \approx 22.67$, times 8 is about 181. The true count is 184, so the estimate is 3 short. Your counts of a square may differ by one or two, and that is fine.

Chapter 4

Multix on deleting. A scientist does not delete readings just because they are awkward. Doing so hides evidence. If a reading is set aside, the reason is written down. **Keep box.** Kept: the typical time and how much the readings varied. Set aside: the order of the readings and which was which.

4.1 Total 65, average 13, spread $15 - 11 = 4$.
4.2 (a) 41. (b) The decimal point was missed; it should be 4.1. (c) $4.1 + 4.0 + 4.2 + 4.1 + 4.1 = 20.5$, and $20.5 \div 5 = 4.1$.
4.3 (a) Both averages are 3.0. (b) P: 0.2, Q: $3.8 - 2.2 = 1.6$. (c) Class P's readings agree better.
4.4 No. Her clock reads about 103 seconds every time, so the average is steady but too long. A steady average shows the result can be repeated, not that the clock is right.
4.5 (a) 25.1 (b) $25.1 \div 8 = 3.1375$ (c) 3.1 (d) $3.3 - 3.0 = 0.3$.
Catch the ruler (practice numbers). Total 130, average 13.0 cm, spread 6 cm, about one sixth of a second.

Chapter 5

Multix on the scale. No. The apple is $152 - 3 = 149$ g. Zero is not just a feeling. **Keep box.** Kept: the corrected length, with the old reading still visible in the wrap. Set aside: the misleading raw reading as an answer.

5.1 A: start 2.0, end 9.4, length 7.4 cm. B: start 1.0, end 12.3, length 11.3 cm. C: start 5.5, end 11.5, length 6.0 cm.
5.2 (a) $205 - 5 = 200$ g. (b) $22.0 - 1.5 = 20.5$ cm.
5.3 C. Reading from the side makes the marks seem to move.
5.4 (a) $400 - 150 = 250$ mL. (b) $35 - 18 = 17$ °C. (c) One reading cannot show whether the jug is steady. It shows only that this jug reads 25 mL above the better jug this once. Read both again and compare before deciding which one is off.
The jug and thermometer figure. 275 mL and 30 °C.

Chapter 6

6.1 (a) O. (b) C. (c) E. (d) K. The ten readings did not test sentence (d). It is a claim about why, and it needs its own test.
6.2 Far too many digits for a ruler marked in millimetres. Write 17.4 cm.
6.3 (a) 123.6 (b) 24.72 (c) 24.7 (d) $24.9 - 24.6 = 0.3$. (e) For example: "The string is about 24.7 cm long; five readings with a mm ruler ranged from 24.6 to 24.9 cm."
6.4 (a) measure many times, average. (b) subtract the zero error. (c) look straight on. (d) round to what the tool can show.

Slop check. (a) You cannot tell: there is no title and no axis labels. (b) Many fonts and colours, stars, rotated filler text and "lol". (c) Axis labels, units, and a title saying what each bar means. (d) The stars, the filler text and the jokes.
Redraw. A good version has one title, labelled axes with units, one or two calm colours, and nothing else.

Whole-book table. Chapter 2: simplified a length by rounding; kept "about how long"; set aside tenths. Chapter 3: simplified a jar by scooping; kept the typical scoop; set aside differences between scoops. Chapter 4: simplified ten readings to an average; kept average and spread; set aside order. Chapter 5: corrected for a start that is not at zero; kept the true length; set aside the misleading raw reading as the answer. Chapter 6: simplified an experiment into five lines; kept the answer with its note; set aside the detail of every step.

Words

average the total of the readings divided by how many there are. Also called the mean.

count a number from counting, such as beans in a scoop.

dot plot a picture with one dot for each reading.

estimate a number about more than was counted, worked out by reasoning.

instrument a tool for measuring, such as a ruler, scale or stopwatch.

middle value the reading in the middle when the readings are put in order. Also called the median.

parallax an apparent change in a reading when you look from the side.

round to replace a number with a nearby, simpler one.

spread the biggest reading minus the smallest.

unit an agreed size to measure with, such as the centimetre.

zero error when an instrument does not start exactly at zero.

Notes for grown-ups

This book teaches that a measurement is a careful judgement, not a perfect fact. Each chapter has one activity at a desk or table, one worked example with made-up practice numbers (labelled as such), one trust box, one "what did we keep?" prompt, and a page of puzzles.

Materials and safety. A ruler marked in millimetres, a second ruler, string, a timer showing seconds, a small cup, a jar, a cup of dried beans or rice, scrap paper. Dried beans are a choking risk for little children and pets. Falling-paper drops are done on the floor, never on furniture. A kitchen scale is optional and needs an adult.

Check boxes. The trust boxes say what a check shows and what it does not show. Repeating a measurement shows that it can be repeated, not that it is right. Two rulers agreeing makes a shared fault unlikely, but does not rule it out.

The practice numbers in the worked examples (beans, falls, pencils) were made up so that the arithmetic is easy to check. They are not results of real experiments, and your own results will differ. That is the point of the book.

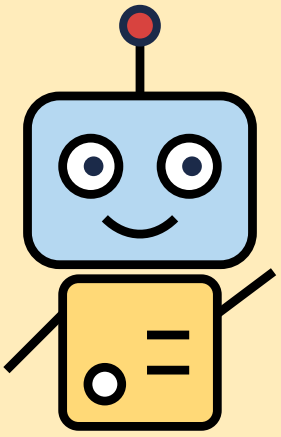
Sources

- Bureau International des Poids et Mesures. *The International System of Units (SI)*, 9th edition, 2019. For the definition of the metre in terms of the speed of light.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For repeated measurement, spread, and the difference between random and systematic error.
- B. N. Taylor and C. E. Kuyatt. *Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results*, NIST Technical Note 1297, 1994. For how to report a result with a stated uncertainty.

Next in Science

A Fair Test

Paper planes fly different distances, and the reason is easy to get wrong. Book 2 shows how to change one thing at a time, keep everything else the same, and write it down so a stranger could repeat your test.



Ages 9 to 11 ⋆ **Science Book 1**